

BUDGET SPLITUP:

I. DEVICE FABRICATION METHODOLOGIES

ZnO/NiO nanomaterial synthesis

As shown in Fig. 2, 80 ml of deionized water is first mixed with 0.5 mmol of zinc acetate dihydrate and 0.25 mmol of nickel acetate tetrahydrate, as well as 0.276 g of SDS and 0.437 g of CTAB. The resulting mixture is then heated for 6 hours at 130°C in a Teflon-lined stainless steel autoclave. The precursor is then calcined for 4 hours at 350°C using a furnace. After cooling naturally, a black color ZnO-NiO composite powder is collected.

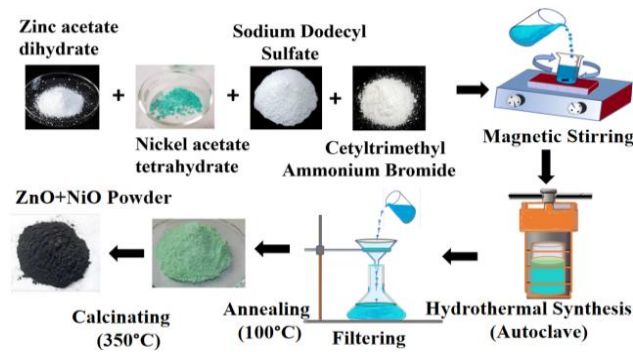


Fig. 2. Hydrothermal Synthesis of SDS:CTAB mixed ZnO/NiO Nanomaterial

Fabrication of the Sensitive Thin Film

The glasses coated with fluorine-doped tin oxide (FTO) are put to use for the thin-film deposition process. The binders used to crush and make the calcined composite powder are ethylcellulose and alpha tripinol. The solution is spin-coated for 10 seconds at 3000 rpm after 10 hours to create a thin sensing layer.

II. MATERIAL CHARACTERIZATION TECHNIQUES

A. X-Ray Diffraction (XRD)

Utilizing XRD, the structure and phase purity of synthesized nanoparticles are evaluated.

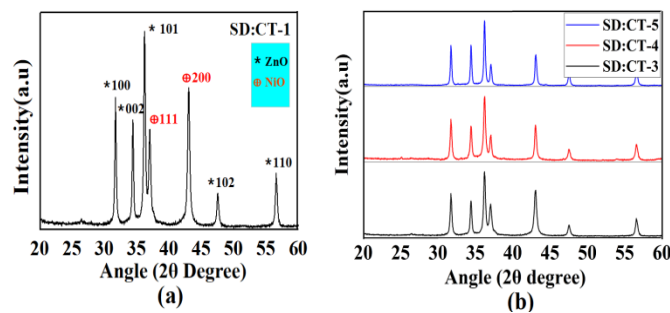


Fig. 3. XRD Pattern of (a) Intensity peaks of SD:CT-1, (b) Integrated pattern Samples

B. Morphological Characterization

FESEM is used to examine the size and morphology of the samples. In Fig. 4, FESEM images of ZnO/NiO at different surfactant molar ratios are displayed.

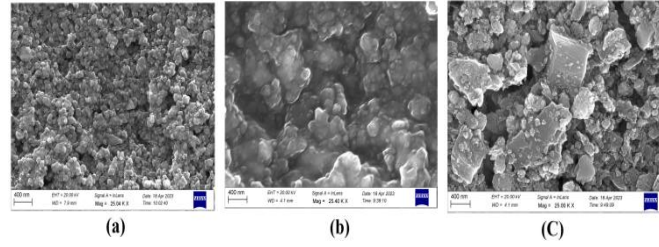


Fig. 4. FESEM image of (a) SD:CT-3 device (b) SD:CT-4 device (c) SD:CT-5 device at 400 nm, 25 K magnification

C. Brunauer- Emmett-Teller (BET) Surface area analysis

The specific surface area (SBET), pore volume, and average diameter Barret-Joyner-Halenda (BJH) of the various concentrations of surfactant mixed ZnO/NiO samples are calculated using nitrogen (N₂) adsorption. The BET equation is given by

$$\frac{1}{W((\frac{P}{P_0})-1)} = \frac{1}{W_m C} + \frac{C-1}{W_m C} (\frac{P}{P_0}) \quad (1)$$

Where, W is the weight of adsorbed gas, $\frac{P}{P_0}$ is the relative pressure, W_m is the weight of adsorbate as a monolayer, and C is the BET surface area constant

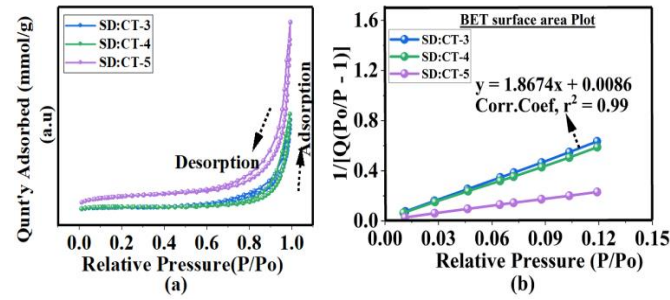


Fig. 5. (a) Nitrogen adsorption-desorption isotherms, (b) BET surface area plot,

D. Current - Voltage characteristics

In the accompanying Fig. 6 (a), (b), and (c), the p-n junction rectifying characteristics of a surfactant-based ZnO/NiO nanosensor are shown, respectively, in the dark and direct sunlight

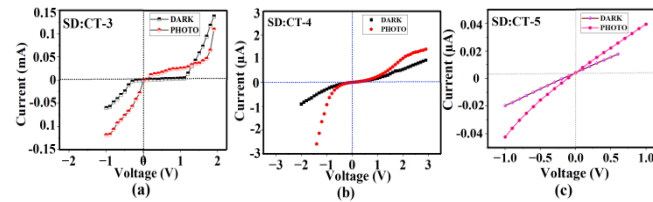


Fig. 6 I-V characteristics of the proposed devices in the dark and photo condition (a) SD:CT-3 device (b) SD:CT-4 device (c) SD:CT-5 device

BILL SPLITUP:

S:No	Description	Qn'ty	Ql'ty	Invoice No. & Date	Receiving Date	Make	Remarks
1	Indicator Papers-Full Range (PH 1.0 to 14.0) 10 BKS Fisher scientific, EthanolLarChinaSqlabuse	2	OK	VS/C2 021/23 -24	27.05.23	---	NIL

2	Sathyabama Institute of Science and Technology	7	OK	2456	12.07.23	---	NIL
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S.No	Description	Qn'ty	Ql'ty	Invoice No. & Date	Receiving Date	Make	Remarks
1	Main components like Arduino, raspberry pi ,bread board	1	OK	Invoice No.: 4537 Date: 29-05-2024	Invoice No.: 4537 Date: 29-05-2024	---	NIL